

# Chuzhditsa: a featural loanword register for Cyrillic, from Bulgarian to the Slavic superset

Author name withheld for review\*

## Abstract

Japanese distinguishes loanwords at a glance: katakana is a parallel register whose visual form alone announces foreignness, and whose adaptation rules map any foreign word to exactly one spelling. No Cyrillic orthography has an equivalent, and the result is systematic indeterminacy (Bulgarian УОШИНГТОН ~ ВОШИНГТОН; Serbian and Russian counterparts abound). We present *Chuzhditsa* (Bulg. чуждица, ‘foreignism’), a three-layer extension of Cyrillic that supplies (i) a featural grammar of diacritic operations deriving letters for non-Slavic phonemes, (ii) a prosodic layer for stress, tone, and length, and (iii) a register-marking typeface whose Glagolitic-derived forms perform the katakana function visually. The letter inventory is assembled conservatively: revivals from historical Bulgarian orthography (the yuses А а) alongside the living Macedonian С с, imports from the wider Cyrillic ecosystem (Kazakh Қ қ, Bashkir Җ җ, Tajik Ў ў), and rule-derived residue. We evaluate the system against the phonologies of twenty major languages, generalize it from its Bulgarian pilot to the full Slavic Cyrillic superset (with deterministic bridges from Latin-alphabet Slavic languages and an external validation on Albanian), and describe a parametric implementation as an OpenType typeface family in which the fusion and anchoring rules of the orthography are executable GSUB/GPOS code. We argue that the laryngeal series (Х х Н н plus the aspiration modifier ʰ) illustrates a general principle for extending alphabets: articulatory gradients should map to visual gradients.

## 1 Introduction

Every literate community borrows words faster than its orthography can naturalize them. Japanese is unusual in having institutionalized the problem: katakana is a full parallel syllabary whose use *is* the marking of foreignness, and whose adaptation conventions ([nart] → ナイト) are deterministic enough that two writers converge on one spelling. The Cyrillic-writing world has no analogous device. Loanwords enter Bulgarian, Russian, Serbian or Ukrainian through ad hoc transcription: sound-based and spelling-based renderings compete (ЛОНЧ ~ ЛЪНЧ for *lunch*), phonemes without Cyrillic letters collapse unpredictably (/θ/ → Т or С; /w/ → В or Y), and nothing in the written form signals that a word is a guest rather than a native.

This paper describes Chuzhditsa, a designed orthographic register that supplies the missing device for Cyrillic, piloted on Bulgarian and then generalized to the Slavic superset. The system makes three claims of general interest to writing-system design. First, that a loanword register can be *featural* in Sampson’s (1985) sense: rather than inventing letters ad libitum, a small set of diacritic operations — each meaning exactly one phonetic feature — derives the entire extension inventory, echoing the compositional logic of Hangul. Second, that such a register can be assembled almost entirely from attested material: dead letters of the target orthography’s own history and living letters of related orthographies, with invention reserved for genuine residue. Third, that the register-marking function that Japanese distributes over a second character inventory can instead be carried by a second *typeface register* — realized here as an OpenType family whose substitution (GSUB) and positioning (GPOS) rules are the orthography’s fusion and attachment rules, executable rather than merely described.

---

\*Fonts, build toolchain, and the twenty-language audit are available in the accompanying repository; the typeface used for all object-language examples in this manuscript is the system being described.

## 2 Background

### 2.1 Katakana as an adaptation layer

Katakana’s success rests on three properties (Irwin 2011; Stanlaw 2004): a fixed sound-mapping table applied to the source word’s pronunciation, not its spelling (silent letters do not survive) — the orthographic reflex of loanword adaptation as phonological mapping (Kang 2011; Peperkamp & Dupoux 2003); graphic distinctness sufficient to mark register at a glance; and extensibility (ヴ for /v/, ファ for /fa/) when the inventory runs out. Chuzhditsa adopts all three as requirements. Katakana also carries a social record that a designed analogue cannot disclaim: loanword marking has been read as a badge of incomplete nativeness, and katakana is deployed to mark foreigners’ accented Japanese as well as foreign words (Irwin 2011; Robertson 2020). We therefore state the register’s semantics narrowly: it asserts that a word’s phonology lies off the native grid — nothing about its right to exist in the language, and nothing about its user. It marks words, never speakers, and a naturalized loan is expected to exit the register, exactly as テンプラ became 天ぷら.

### 2.2 The Cyrillic commonwealth

Cyrillic serves some fifty languages (Comrie 1996), and the non-Slavic ones have already engineered letters for most sounds Slavic lacks: Kazakh қ /q/, ң /ŋ/, ғ /ɣ/; Bashkir җ /θ/, ҙ /ð/; Tajik ӣ /i:/ and ҳ /h/ in Tajik, reassigned here to /h/ — the Caucasian digraph xI is the nearer precedent); Altai ӧ /ø/, ӧ̈ /y/; Belarusian ь /w/; Serbian ɟ /dʒ/; the Caucasian palochka I. These letters carry decades of typographic and pedagogic practice. A design that imports them inherits that practice for free — an option unavailable to katakana’s designers and, we will argue, the single largest economy in the present system.

### 2.3 The Glagolitic precedent

Bulgaria is the locus of a script replacement: Glagolitic (863 CE), the first Slavic alphabet, was gradually displaced by Cyrillic (Preslav Literary School, c. 893), which modelled most letters on Greek uncial and drew on Glagolitic mainly for the sounds Greek lacked (Schenker 1995; Cubberley 1996). Glagolitic then survived for a millennium as a special-register script (Croatian liturgical use into the twentieth century; the last Glagolitic missal was printed in 1905). Chuzhditsa knowingly reverses the vector: the visual DNA of the *oldest* Slavic letterforms — round bowls, unicameral proportions — marks the *newest* words, and the letters it revives (А А А А) inherit the sound-slots of Glagolitic ancestors (the Cyrillic shapes themselves are reworkings).

## 3 Design principles

1. **Determinism.** One phoneme, one rendering; two transcribers converge. Loss is permitted but must be declared (§9).
2. **Featural transparency.** Every diacritic operation means exactly one feature and applies wherever the feature appears.
3. **Pair symmetry.** Obstruent letters enter the system in voiceless/voiced pairs aligned with the native series. This principle outranks heritage: the historical theta-letter fita (Θ) was rejected because it has no voiced partner and derives from no rule; the Bashkir pair җ/ҙ (/θ//ð/) does its work systematically.
4. **Precedent first.** Revive native dead letters, then import living Cyrillic letters, and only then derive new combinations; never invent an unattested base shape.

5. **Pan-Cyrillic legibility.** Every base skeleton follows the reader’s existing Cyrillic, and every extension must be guessable at first sight by any Cyrillic reader from its base letter plus a transparent mark — a design criterion the orthography-development literature calls transferability (Cahill & Rice 2014).
6. **Register by typeface.** Foreignness is marked by letterform register, not by punctuation or a second inventory: text set in the Chuzhditsa face is thereby marked, exactly as katakana text is — a register distinction of the kind the biscriptality literature treats as script-internal diglossia (Sebba 2007; Bunčić, Lippert & Rabus 2016).

## 4 The featural grammar

Table 1 gives the full operation set. The retraction hook generalizes a precedent internal to the Slavic base inventory itself —  $\text{ш}$  is  $\text{ш}$  with a hook — and the glide breve generalizes  $\text{и} \rightarrow \text{й}$ . Where Unicode supplies precomposed hooked letters ( $\text{ѡ}$   $\text{ѣ}$   $\text{ѧ}$   $\text{Ѩ}$   $\text{Ѫ}$ ) they are used; elsewhere the same hook is written with combining U+0322 ( $\text{Ѣ}$   $\text{Ѥ}$   $\text{Ѧ}$ ).

| Mark           | Operation                         | Derivations                                                                                                                           |
|----------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| ◌̆ breve       | vowel $\rightarrow$ glide         | $\text{и} \rightarrow \text{й}$ /j/ · $\text{γ} \rightarrow \text{ѣ}$ /w/                                                             |
| ◌̇ hook below  | place shifted off the native grid | $\text{ѡ}$ /q/ · $\text{ѣ}$ /h/ · $\text{ѧ}$ /ŋ/ · $\text{Ѩ}$ /θ/ · $\text{Ѫ}$ /ð/ · $\text{Ѣ}$ /t/ · $\text{Ѥ}$ /d/ · $\text{Ѧ}$ /t/ |
| ◌̈ crossbar    | lenition                          | $\text{Г} \rightarrow \text{Ѧ}$ /x/                                                                                                   |
| ◌̊ double dot  | vowel fronting                    | $\text{Ӑ}$ /æ/ · $\text{Ӓ}$ /ø œ/ · $\text{Ӕ}$ /y ʏ/                                                                                  |
| ◌̋ yus tail    | nasalization                      | $\text{Ӑ}$ /õ/ · $\text{Ӓ}$ /ẽ/ · $\text{Ӕ}$ /ǎ/ · $\text{Ӧ}$ /œ/                                                                     |
| ◌̌ ◌̍ raised h | aspiration; breathy voice         | $\text{п̌}$ $\text{т̌}$ $\text{к̌}$ $\text{ч̌}$ · $\text{б̌}$ $\text{а̌}$ $\text{г̌}$ $\text{у̌}$                                     |
| ◌̄ macron      | vowel length                      | $\text{й̄}$ $\text{ѣ̄}$ (precomposed) · $\text{Ӑ̄}$ $\text{Ӓ̄}$ $\text{Ӕ̄}$                                                           |
| ◌̇ palochka    | laryngeal–pharyngeal action       | $\text{т̇}$ $\text{к̇}$ $\text{ч̇}$ (ejectives) · bare $\text{I} = \text{/ʕ/}$                                                        |
| ◌̈́            | palatalization                    | $\text{ѡ̈́}$ /j/ · $\text{ѣ̈́}$ /ɟ/ · $\text{ѧ̈́}$ /ç/ · $\text{Ѩ̈́}$ /ç/                                                             |

Table 1: The featural operations. Each mark has one meaning; precomposed Unicode letters are used where they exist.

### 4.1 The laryngeal gradient

The three back fricatives that plague Cyrillic transcription of Arabic, English and German — /x/, /ħ/, /h/ — receive a designed gradient: articulatory weakening maps to graphic lightening. Native  $\text{x}$  (/x/~/χ/) is two full crossing strokes, the densest friction; pharyngeal  $\text{ħ}$  (/ħ/) is  $\text{x}$  pulled deeper by the retraction hook; glottal  $\text{h}$  (/h/~/ɦ/) is a single open arc, literally the most open glyph of the three; and aspiration  $\text{h}^h$ , a feature rather than a segment, is  $\text{h}$  reduced to a superscript. Arabic exhibits the full triple within ordinary vocabulary:  $\text{هـ}$  /hila:l/,  $\text{مُحَمَّد}$  /muħammad/,  $\text{خَالِد}$  /xa:lɪd/. We suggest the mapping *articulatory gradient*  $\rightarrow$  *visual gradient* as a reusable principle for alphabet extension.

### 4.2 The Indic stress test

Hindi–Urdu and Bengali cross a four-way laryngeal contrast with a dental/retroflex place contrast. The grammar covers the entire grid with zero new letters:  $\text{т}$  /t/,  $\text{т̌}$  /ť/,  $\text{Ӑ}$  /d/,  $\text{Ӑ̌}$  /ď/; retroflex  $\text{ṭ}$  /ṭ̌/  $\text{Ṭ}$  /Ṭ̌/  $\text{Ṭ̌}$  /Ṭ̌/, plus  $\text{ɦ}$  /ɦ/,  $\text{ṣ}$  /ṣ/,  $\text{ṣ̌}$  /ṣ̌/. A single word such as  $\text{तृणक}$  (*ṭhīk*, /ṭ̌i:k/) composes three features on one base — hook, aspiration, length — which is precisely the featural claim made concrete.



As an external stress test we applied the system to Albanian, a non-Slavic Balkan language: every Albanian phoneme lands on an existing letter ( $\text{ë } /ə/ \rightarrow \text{ѐ}$ ,  $th/dh \rightarrow \text{џ}$ ,  $y \rightarrow \text{ѣ}$ ,  $xh \rightarrow \text{ѡ}$ ,  $x \text{ /dz/} \rightarrow \text{Ѣ}$ ,  $q/gj \rightarrow \text{кѣ/гѣ}$ ): Шкѣипѣриа, Тирѣнѣ. A system built for Slavic phonology plus the featural operations appears to cover the Balkan sprachbund without amendment. The practical beneficiary is not Albanian itself, which is securely Latin-written, but Cyrillic-language text citing Albanian: the  $\text{ë}$  that Macedonian and Serbian renderings currently drop is exactly  $\text{ѐ}$ .

## 8 Typeface implementation

The Chuzhditsa family (Regular, Bold, Italic, Bold Italic; TTF and CFF/OTF) is generated by a  $\sim 300$ -line parametric builder: every glyph is a set of monoline strokes — lines, arcs, and the full circles that give the face its Glagolitic character — expanded to outlines by a union of per-segment rectangles and per-vertex discs, with weight and slant as parameters. Capitals stand at 700 units, lowercase at an x-height of 500 with true two-case architecture (bowl-and-stem  $\text{а}$ , ascending  $\text{б}$ , descending  $\text{р}$ ). Every combining mark receives computed GPOS mark-to-base anchors (the macron of  $\text{ā}$  attaches 251 units lower than that of  $\text{Ā}$ ; the retroflex hook of  $\text{ṛ}$  lands on the letter’s stem), and the ligature stratum is a GSUB *liga* feature — the orthography’s rules are the font’s code, verified by HarfBuzz shaping tests in the build toolchain — a concrete instance of Sproat’s (2000) treatment of orthography as a computable mapping. The manuscript the reader is holding sets all object-language material in this typeface.

## 9 Evaluation: the twenty-language audit

The system was audited against the phoneme inventories of the twenty most-spoken languages (clicks excluded by scope). Table 2 samples the audit; the full 126-item set accompanies the repository. Declared mergers are: all rhotics  $\rightarrow \text{р}$ ;  $/s/ \text{ /ʃ/} \rightarrow \text{ш}$ ;  $/\phi/ \text{ /f/} \rightarrow \text{ф}$ ;  $/h/ \text{ /ɦ/} \rightarrow \text{н}$ ; implosives  $\rightarrow \text{б а}$ ;  $/q/ \text{ /g/} \rightarrow \text{к}$ ; Korean tense consonants as geminates; English input referenced to a rhotic accent (GA);  $/r/$  dropped outside Arabic material.

| Source   |                      | IPA         | Chuzhditsa |
|----------|----------------------|-------------|------------|
| English  | <i>thanks</i>        | [θæŋks]     | џАѢКС      |
| English  | <i>weekend</i>       | [ˈwiːkɛnd]  | ѡѢКЕНА     |
| Mandarin | 北京 <i>Běijīng</i>    | [pèi.tɕíŋ]  | ПѢЙЧИѢ     |
| Hindi    | ठीक <i>thīk</i>      | [tʰiːk]     | ṛṛṛѢК      |
| Arabic   | محمد <i>Muḥammad</i> | [muˈħammad] | МухАММАД   |
| French   | <i>croissant</i>     | [kʁwasɑ̃]   | крѡасѢ     |
| French   | <i>bien</i>          | [bjɛ̃]      | БѢ         |
| Polish   | <i>Wrocław</i>       | [ˈvrɔʦwaf]  | ВроџѡѢф    |
| Japanese | 東京 <i>Tōkyō</i>      | [toːkʰoː]   | ТѡкѡѢ      |
| Korean   | 한글 <i>hangul</i>     | [hangwɭ]    | ҺАнГѡЛ     |
| Georgian | ყვალის <i>c'qali</i> | [tʰsˈqʰali] | џѢṼАЛИ     |
| Albanian | <i>Shqipëria</i>     | [ʃcipəˈria] | Шкѣипѣриа  |

Table 2: Sampled audit items; the Chuzhditsa column is set in the typeface under discussion, with live GSUB fusion and GPOS anchoring.

## 10 Limitations

The retraction hook is polysemous by direction — uvular on  $\text{к}$ , pharyngeal on  $\text{х}$ , interdental on  $\text{с ѣ}$ , retroflex on  $\text{т а р}$ . No attested language contrasts two of its values on one base letter, so the ambiguity

is theoretical, but it is a real weakening of the one-mark-one-meaning claim. /ɕ/ keeps two spellings (ɕ under contrast, ɕ by entrenchment) — the one point where usage beat system. Six-tone prosodies fold lossily into four marks. Click phonology is future work, as is a mark-to-mark (mkmk) positioning pass for the rare double-stacked vowel.

## 11 Conclusion

Chuzhditsa demonstrates that the katakana function — deterministic adaptation plus visible register — can be retrofitted onto an alphabet without inventing a parallel inventory: a featural grammar over attested Cyrillic material covers twenty phonologies, one historical script supplies the register aesthetics, and the whole specification compiles to a font whose OpenType rules *are* the orthography. The generalization from Bulgarian to the Slavic superset cost one letterform (e) and no grammar changes, which we take as evidence that the featural layer, not the letter list, is the right unit of design.

## References

- [1] Bunčić, D., S. L. Lippert & A. Rabus (eds.). 2016. *Biscriptality: A Sociolinguistic Typology*. Heidelberg: Universitätsverlag Winter.
- [2] Cahill, M. & K. Rice (eds.). 2014. *Developing Orthographies for Unwritten Languages*. Dallas, TX: SIL International.
- [3] Comrie, B. & G. G. Corbett (eds.). 1993. *The Slavonic Languages*. London: Routledge.
- [4] Comrie, B. 1996. Adaptations of the Cyrillic alphabet. In P. Daniels & W. Bright (eds.), *The World's Writing Systems*, 700–726. New York: OUP.
- [5] Coulmas, F. 2003. *Writing Systems: An Introduction to their Linguistic Analysis*. Cambridge: CUP.
- [6] Cubberley, P. 1996. The Slavic alphabets. In P. Daniels & W. Bright (eds.), *The World's Writing Systems*, 346–355. New York: OUP.
- [7] Данчев, А. 1982. *Българска транскрипция на английски имена*. 2. изд. София: Народна просвета.
- [8] Daniels, P. & W. Bright (eds.). 1996. *The World's Writing Systems*. New York: OUP.
- [9] Irwin, M. 2011. *Loanwords in Japanese*. Amsterdam/Philadelphia: John Benjamins.
- [10] Kang, Y. 2011. Loanword phonology. In M. van Oostendorp, C. J. Ewen, E. Hume & K. Rice (eds.), *The Blackwell Companion to Phonology*, vol. 4, 2258–2282. Malden, MA: Wiley-Blackwell.
- [11] Peperkamp, S. & E. Dupoux. 2003. Reinterpreting loanword adaptations: The role of perception. *Proceedings of the 15th International Congress of Phonetic Sciences*, 367–370.
- [12] Robertson, W. C. 2020. *Scripting Japan: Orthography, Variation, and the Creation of Meaning in Written Japanese*. London: Routledge.
- [13] Sampson, G. 1985. *Writing Systems: A Linguistic Introduction*. Stanford: Stanford University Press.
- [14] Sebba, M. 2007. *Spelling and Society: The Culture and Politics of Orthography around the World*. Cambridge: CUP.

- [15] Schenker, A. M. 1995. *The Dawn of Slavic: An Introduction to Slavic Philology*. New Haven: Yale University Press.
- [16] Sproat, R. 2000. *A Computational Theory of Writing Systems*. Cambridge: CUP.
- [17] Stanlaw, J. 2004. *Japanese English: Language and Culture Contact*. Hong Kong: Hong Kong University Press.
- [18] The Unicode Consortium. 2024. *The Unicode Standard*, Version 16.0.0. South San Francisco, CA: The Unicode Consortium.